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WHEELCHAIR TRANSPORT CARRIER

Abstract

A wheelchair transport carrier and method. In an embodiment, a wheelchair transport carrier comprises a main frame comprising a support platform defining a wheelchair support surface to support wheels of a wheelchair, a plurality of lift handles to lift the main frame, and a plurality of restraints to anchor the wheelchair to the support platform.

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Background/Summary

RELATED APPLICATIONS [0001] This non-provisional application claims priority to U.S. provisional application 63/562,163, filed on Mar. 6, 2024, which is incorporated herein by reference as if fully provided herein.

FIELD

[0002] This disclosure is related to the field of wheelchairs, and more particularly, to transport of wheelchairs.

BACKGROUND

[0003] People who cannot walk or have difficulty walking may use a wheelchair as a means of mobility. A problem may arise when traveling, such as on an airline, as the wheelchair may be damaged due to improper handling by airline personnel, improper storage during transport, etc. For example, electric or powered wheelchairs may be hundreds of pounds, and are difficult to lift onto conveyors, into cargo holds, etc. Thus, the wheelchair may tip over, fall from an elevated surface (e.g., conveyor), or otherwise be damaged. Further, there is no clear indication where it is safe to lift some types of wheelchairs, which results in armrests and/or other components being damaged during lifting. Damage to a wheelchair is not only inconvenient to the owner, but may be their only means of mobility at their destination. Further, airlines and/or other types of public or private transport organizations are typically liable for repair or replacement of damaged wheelchairs.

SUMMARY

[0004] The present embodiments describe a transport carrier for wheelchairs. Although the description below may focus on wheelchairs, the transport carrier may be used for scooters and/or other types of conveyances. In general, the transport carrier comprises a support platform that supports the wheels of the wheelchair, and a plurality of restraints that anchor the wheelchair to the support platform. The transport carrier further comprises a plurality of lift handles comprising lift points for the transport carrier. When loaded on the transport carrier and restrained, the wheelchair may be properly handled and stored during transport to avoid damage to the wheelchair. One technical benefit is people with disabilities or that otherwise use a wheelchair feel freer to travel without worry of their wheelchair being damaged. Another technical benefit is airlines and/or other organizations can avoid liabilities associated with damaged wheelchairs.

[0005] In an embodiment, a wheelchair transport carrier comprises a main frame comprising a support platform defining a wheelchair support surface to support wheels of a wheelchair, a plurality of lift handles to lift the main frame, and a plurality of restraints to anchor the wheelchair to the support platform.

[0006] In an embodiment, a wheelchair transport carrier comprises a main frame comprising a support platform defining a wheelchair support surface to support wheels of a wheelchair, a ramp at a loading end of the main frame configured to rotate in relation to the main frame between a raised position and a lowered position, a plurality of restraints to anchor the wheelchair to the support platform, and a plurality of lift handles rotatably coupled to the main frame to transition between a stowed position and an open position.

[0007] In an embodiment, a method comprises obtaining a wheelchair transport carrier, lowering the ramp to the lowered position, loading the wheelchair on the wheelchair support surface via the ramp, restraining the wheelchair to the main frame via one or more restraints, raising the ramp to the raised position, opening the lift handles to the open position, and lifting the wheelchair transport carrier via the lift handles with the wheelchair restrained.

[0008] The above summary provides a basic understanding of some aspects of the specification. This summary is not an extensive overview of the specification. It is intended to neither identify key or critical elements of the specification nor delineate any scope particular embodiments of the specification, or any scope of the claims. Its sole purpose is to present some concepts of the

specification in a simplified form as a prelude to the more detailed description that is presented later.

Description

DESCRIPTION OF THE DRAWINGS

[0009] Some embodiments are now described, by way of example only, and with reference to the accompanying drawings. The same reference number represents the same element or the same type of element on all drawings.

[0010] FIGS. **1-4** are perspective views of a wheelchair transport carrier in illustrative embodiments.

[0011] FIG. **5** is a flow chart illustrating a method of using a wheelchair transport carrier in an illustrative embodiment.

[0012] FIGS. **6-7** are perspective views of a wheelchair transport carrier in an illustrative embodiment.

[0013] FIG. **8** is a plan view of a wheelchair transport carrier in an illustrative embodiment.

[0014] FIG. **9** is a side view of a wheelchair transport carrier in an illustrative embodiment.

[0015] FIG. **10** is an end view of a wheelchair transport carrier in an illustrative embodiment.

[0016] FIG. **11** is an end view of a wheelchair transport carrier in an illustrative embodiment.

[0017] FIGS. **12A-12B** are perspective views of a wheelchair transport carrier in an illustrative embodiment.

[0018] FIG. **13** illustrates a ramp latch in an illustrative embodiment.

[0019] FIG. **14** illustrates a wheelchair loaded onto a wheelchair transport carrier.

[0020] FIG. **15** is a plan view of a wheelchair transport carrier in an illustrative embodiment.

[0021] FIG. **16** illustrates a restraint in an illustrative embodiment.

[0022] FIGS. **17A-17F** illustrate a lift handle in illustrative embodiments.

[0023] FIG. **18** is a perspective view of a wheelchair transport carrier in an illustrative embodiment.

[0024] FIGS. **19A-19B** are flow charts illustrating a method of using a wheelchair transport carrier in an illustrative embodiment.

DESCRIPTION OF EMBODIMENTS

[0025] The figures and the following description illustrate specific exemplary embodiments. It will thus be appreciated that those skilled in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the embodiments and are included within the scope of the embodiments. Furthermore, any examples described herein are intended to aid in understanding the principles of the embodiments, and are to be construed as being without limitation to such specifically recited examples and conditions. As a result, the inventive concept(s) is not limited to the specific embodiments or examples described below, but by the claims and their equivalents.

[0026] FIG. **1** is a perspective view of a wheelchair transport carrier **100** in an illustrative embodiment. FIG. **1** is not drawn to scale. Wheelchair transport carrier **100** comprises an apparatus, equipment, or other structure configured to secure a wheelchair during transport, such as on an airline. In an embodiment, wheelchair transport carrier **100** includes a main frame **102** having a top side **104** and an opposing bottom side **106** (also referred to as an undercarriage). Main frame **102** may be made from a rigid material, such as aluminum, carbon fiber, or another material. Top side **104** of main frame **102** comprises a support platform **107** that defines a wheelchair support surface **108** configured to support wheels of a wheelchair (not shown). In an embodiment, wheelchair support surface **108** forms a generally flat or planar surface upon which wheels of a wheelchair may be loaded.

[0027] In an embodiment, main frame **102** may be a rectangular or elongated member having a length **132** (e.g., in the y-direction) and a width **130** (e.g., in the x-direction). The dimensions of wheelchair transport carrier **100** may be configured to fit within the door of a cargo hold of an aircraft or another transportation vehicle. For example, the width **130** of wheelchair transport carrier **100** may be less than about 30 inches, less than about 31 inches, less than about 32 inches, less than about 36 inches, etc., the length **132** of wheelchair transport carrier **100** may be less than about 48 inches, less than about 50 inches, etc.

[0028] On or proximate to wheelchair support surface **108** of main frame **102**, wheelchair transport carrier **100** comprises a plurality of restraints **110** configured to anchor the wheelchair to the support platform **107** of main frame **102**. Main frame **102** may comprise a plurality of anchor points **120** where restraints **110** are affixed or attached to main frame **102**. In an embodiment, restraints **110** may comprise automatic retractable tie-downs, such as Q-strains. The restraints **110** may include a hook end **112** configured to attach to the wheelchair, and an anchor end that is affixed or attached to an anchor point **120** of main frame **102**. One technical benefit is a wheelchair may be securely attached to wheelchair transport carrier **100** during transport.

[0029] FIG. **2** is a perspective view of wheelchair transport carrier **100** in an illustrative embodiment. FIG. **2** is not drawn to scale. Wheelchair transport carrier **100** further comprises a plurality of lift handles **212** configured to lift the main frame **102**. As shown in FIG. **2**, main frame **102** comprises opposing ends **202-203**, and opposing sides **204-205**. Sides **204-205** of main frame **102** may include raised side panels **210** that are raised or elevated in relation to the wheelchair support surface **108**. Lift handles **212** may be disposed on or at side panels **210** of main frame **102** to provide a lifting point for wheelchair transport carrier **100**. However, the lift handles **212** may be disposed at other locations on main frame **102** in other embodiments. In an embodiment, lift handles **212** may be extendible or retractable, foldable, etc. One technical benefit is the lift handles **212** provide safe locations from which to lift or move the wheelchair transport carrier **100**. Thus, a restrained wheelchair does not need to be contacted or handled when lifting or moving the wheelchair transport carrier **100**.

[0030] FIG. **3** is a perspective view of wheelchair transport carrier **100** in an illustrative embodiment. In an embodiment, wheelchair transport carrier **100** may further comprise a ramp **302** disposed at an end **202** of the main frame **102**, which may be referred to as a loading end. Ramp **302** is configured to facilitate loading of a wheelchair onto wheelchair support surface **108** at loading end **202**. In an embodiment, ramp **302** may be retractable, foldable, etc. One technical benefit is a wheelchair may be safely loaded onto wheelchair transport carrier **100**, without having to physically lift the wheelchair itself.

[0031] FIG. **4** is a perspective view of wheelchair transport carrier **100** in an illustrative embodiment. In an embodiment, wheelchair transport carrier **100** may further comprise one or more wheels **404** or rollers mounted on bottom side **106** of main frame **102** toward end **203**. Wheelchair transport carrier **100** may further comprise one or more pull handles **402** mounted toward end **202** of main frame **102** opposite end **203**. Pull handles **402** are configured to pull or drag wheelchair transport carrier **100** on wheels **404** along a ground surface. In an embodiment, pull handles **402** may be retractable, foldable, etc. One technical benefit is the portability of the wheelchair transport carrier **100** is enhanced or improved.

[0032] In an embodiment, anchor points **120** may comprise L-track rails **406** or another type of track system configured to adjust the positions/locations of restraints **110**. One technical benefit is the positions/locations of restraints **110** are adjustable to accommodate different wheelchairs.

[0033] FIG. **5** is a flow chart illustrating a method **500** of using a wheelchair transport carrier **100** in an illustrative embodiment. The steps of the flow charts described herein are not all inclusive and may include other steps not shown, and the steps may be performed in an alternative order. For method **500**, a wheelchair transport carrier **100** is obtained (step **502**), such as described above. A wheelchair is loaded on the wheelchair support surface **108** of the wheelchair transport carrier **100**

(step 504), such as via a ramp 302. The wheelchair is restrained or anchored to the wheelchair transport carrier 100 via the restraints 110 (step 506). The wheelchair transport carrier 100 may then be lifted via the lift handles 212 with the wheelchair restrained (step 508), such as onto an aircraft, onto a conveyor, etc. One technical benefit is a wheelchair may be protected from damage during transport, such as on an aircraft.

[0034] Although specific embodiments were shown above, wheelchair transport carrier 100 may include further structures or features that assist in the transport of a wheelchair, assist in loading a wheelchair, assisting in lifting the wheelchair transport carrier 100 with a wheelchair restrained, etc. In the following examples, additional or alternative structures or features may be described in the context of a wheelchair transport carrier, which provide similar benefits as described above. These examples may be incorporated in embodiments described above as desired.

[0035] FIGS. 6-7 are perspective views of a wheelchair transport carrier 600 in an illustrative embodiment. In an embodiment, wheelchair transport carrier 600 includes a main frame 602 made from a rigid material, such as aluminum, carbon fiber, and/or other materials. Main frame 602 comprises a support platform 607 having a top side 604 and an opposing bottom side 606 (also referred to as an undercarriage). Top side 604 of support platform 607 comprises or defines a wheelchair support surface 608 configured to support wheels of a wheelchair (not shown). In an embodiment, wheelchair support surface 608 forms a generally flat or planar surface upon which wheels of a wheelchair may be loaded. The bottom side 606 (or at least a portion thereof) is configured to rest on a support surface, such as the ground, a floor, etc.

[0036] FIG. 8 is a plan view of wheelchair transport carrier 600 in an illustrative embodiment. In an embodiment, wheelchair transport carrier 600/main frame 602/support platform 607 may be a rectangular or elongated member having a length 830 (e.g., in the y-direction) and a width 832 (e.g., in the x-direction). The dimensions of wheelchair transport carrier 600 may be configured to fit within the door of a cargo hold of an aircraft or another transportation vehicle. For example, the width 832 of wheelchair transport carrier 600 may be less than about 30 inches, less than about 31 inches, less than about 32 inches, less than about 36 inches, etc., the length 830 of wheelchair transport carrier 600 may be less than about 48 inches, less than about 50 inches, etc.

[0037] In FIGS. 6-7, main frame 602 may further comprise side rails 610 (also referred to as side panels) connected or affixed to support platform 607. Main frame 602 comprises opposing ends 622-623, and opposing sides 624-625. Side rails 610 are connected or affixed to support platform 607 at sides 624-625, such as by bolts or other fastening devices, welds, etc., and are raised or elevated in relation to the wheelchair support surface 608. FIG. 9 is a side view of wheelchair transport carrier 600 in an illustrative embodiment. In an embodiment, side rails 610 may comprise one or more horizontal rail members 616 that span at least a portion of the length 830 of the main frame 602. The side rails 610 may further comprise a plurality of vertical rail members 614 that connect the horizontal rail member(s) 616 to the support platform 607. The vertical rail members 614 may be referred to as support posts, support bars, stanchions, etc. One or more of vertical rail members 614 may be perpendicular to the plane of wheelchair support surface 608. Although the term “vertical” is used to describe rail members 614, one or more of the vertical rail members 614 may be angled in relation to the plane of wheelchair support surface 608. The horizontal rail member(s) 616 and vertical rail members 614 may comprise square or rectangular tubing (e.g., hollow), cylindrical tubing, etc.

[0038] In FIGS. 6-7, wheelchair transport carrier 600 may further comprise an end rail 611 (also referred to as an end panel) connected or affixed to support platform 607 and/or side rails 610. End rail 611 is connected or affixed at end 623, such as by bolts or other fastening devices, welds, etc. FIG. 10 is an end view of wheelchair transport carrier 600 in an illustrative embodiment. In an embodiment, end rail 611 may comprise one or more horizontal rail members 616 that span at least a portion of the width 832 of the main frame 602. The horizontal rail member 616 of end rail 611 may be connected or affixed to horizontal rail members 616 of side rails 610. In other embodiments,

end rail **611** may connect to support platform **607** through one or more vertical rail members **614**. One technical benefit is side rails **610** and/or end rail **611** surround the wheelchair support surface **608** on sides **624-625** and end **623** to form an enclosure that may further protect a restrained wheelchair.

[0039] In an embodiment, wheelchair transport carrier **600** may further comprise one or more wheels **680** or rollers mounted on support platform **607** toward end **623**. One technical benefit is the portability of the wheelchair transport carrier **600** is enhanced or improved. In an embodiment, wheelchair transport carrier **600** may further comprise a ramp **662** disposed at end **622** of the main frame **602**, which may be referred to as a loading end, as illustrated in FIGS. **6-7**. Ramp **662** is configured to facilitate loading of a wheelchair onto wheelchair support surface **608** of main frame **602** at loading end **622**. FIG. **11** is an end view of wheelchair transport carrier **600** in an illustrative embodiment. FIGS. **12A-12B** are perspective views of wheelchair transport carrier **600** in an illustrative embodiment. Ramp **662** is attached or affixed to main frame **602** (e.g., support platform **607**) at end **622** via a ramp hinge **1106**. Ramp hinge **1106** allows ramp **662** to rotate or pivot in relation to main frame **602** about a rotation axis **1210** (see FIGS. **12A-12B**), which is generally a horizontal axis. Ramp **662** is configured to rotate via ramp hinge **1106**, and transition between a closed position **1202** (or closed configuration, raised position, etc.) as shown in FIG. **12A**, and an open position **1204** (or open/loading configuration, lowered position, etc.) as shown in FIG. **12B**. When in the closed position **1202**, the ramp **662** may be approximately vertical in relation to the ground, the wheelchair support surface **608**, etc. Wheelchair transport carrier **600** may comprise one or more ramp latches **1104** configured to lock or secure the ramp **662** in the closed position **1202**. FIG. **13** illustrates a ramp latch **1104** in an illustrative embodiment. The ramp latch **1104** may comprise a slide bolt latch **1304** (or slide bolt lock). Slide bolt latch **1304** includes a (e.g., metal) bolt **1306** or barrel configured to slide into a receptacle or strike plate on main frame **602**, such as on a horizontal rail member **616**, a vertical rail member **614**, etc.

[0040] FIG. **12B** illustrates the ramp **662** in an open position **1204**. The inner surface **1206** of ramp **662** may be a generally flat or planar surface that is inclined from a ground surface to the wheelchair support surface **608**. This allows for a wheelchair to roll from the ground surface, up the ramp **662**, and onto the wheelchair support surface **608**. FIG. **14** illustrates a wheelchair **1402** loaded onto wheelchair transport carrier **600**. Although a non-electric wheelchair is illustrated in FIG. **14** merely as an example, wheelchair transport carrier **600** may be used for transport of electric wheelchairs or other electric conveyances that are relatively heavy.

[0041] Although ramp **662** is shown as being foldable in the above embodiment, ramp **662** may be retractable in other embodiments. One technical benefit is a wheelchair **1402** may be safely loaded onto wheelchair transport carrier **600**, without having to physically lift the wheelchair **1402** itself.

[0042] With a wheelchair **1402** loaded onto wheelchair transport carrier **600**, the wheelchair **1402** may be restrained onto wheelchair transport carrier **600**. FIG. **15** is a plan view of wheelchair transport carrier **600** in an illustrative embodiment. Wheelchair transport carrier **600** may comprise a plurality of anchor points **1510** where restraints are affixed or attached to main frame **602**. The restraints are configured to anchor the wheelchair **1402** to the main frame **602**. For example, anchor points **1510** may be positioned at or near corners of wheelchair support surface **608**. In an embodiment, anchor points **1510** may comprise L-track rails **406** or another type of track system configured to adjust the positions/locations of the restraints. One technical benefit is the positions/locations of restraints are adjustable to accommodate different wheelchairs **1402**.

[0043] FIG. **16** illustrates a restraint **1610** in an illustrative embodiment. In an embodiment, a restraint **1610** may include a hook end **1612** configured to attach to a wheelchair **1402**, and an anchor end **1614** that is affixed or attached to an anchor point **1510** of main frame **602**. For example, the anchor end **1614** may be attached to an L-track rail **406** installed on wheelchair support surface **608**. Restraint **1610** further includes a strap **1616** and a tensioner mechanism **1618** (e.g., automatic tensioner) between the hook end **1612** and the anchor end **1614**. Restraint **1610**

may be referred to as an automatic retractable tie-down **1620**, one example of which is a Q-strait. One technical benefit is a wheelchair **1402** may be securely attached to wheelchair transport carrier **600** during transport.

[0044] In FIGS. **6-7**, wheelchair transport carrier **600** further comprises a plurality of lift handles **612**. Lift handles **612** are structural elements configured to provide lift points to lift the main frame **602**. Lift handles **612** may be disposed at the sides **624-625** and/or corners **671-674** of main frame **602**. Although four lift handles **612** are illustrated, there may be more or less lift handles **612** in other embodiments. One technical benefit is the lift handles **612** provide safe locations from which to lift or move the wheelchair transport carrier **600**. Thus, a restrained wheelchair **1402** does not need to be contacted or handled when lifting or moving the wheelchair transport carrier **600**.

[0045] FIGS. **17A-17F** illustrate a lift handle **612** in illustrative embodiments. In an embodiment, a lift handle **612** may be rotatably or pivotally coupled to main frame **602** to transition between a closed position **1710** (see FIG. **17A**) and an open position **1712** (see FIG. **17F**). In an embodiment, a lift handle **612** may be rotatably or pivotally coupled to main frame **602**. As in FIG. **17A**, a vertical rail member **614** may comprise a round tube, and lift handle **612** may be configured to slide over the vertical rail member **614** and rotate in relation to the vertical rail member **614** about a rotation axis **1770**, which may be generally a vertical axis. A lift handle **612** may comprise a handle member **1702**, a rotation locking mechanism **1704** (also referred to as rotation locking member), and a biasing member **1706**. Handle member **1702** is configured to rotate about rotation axis **1770** between a closed position **1710** (also referred to as stowed or retracted position, a closed configuration, etc.) as shown in FIG. **17A**, and an open position **1712** (also referred to as an extended position or lifting position, an open configuration, etc.) as shown in FIG. **17F**. Rotation locking mechanism **1704** is configured to prevent rotation of the handle member **1702**. For example, rotation locking mechanism **1704** is configured to engage a corresponding rotation locking mechanism **1750** (also referred to as rotation locking member) of main frame **602** to lock or secure the handle member **1702** in the closed position **1710** or the open position **1712**. Biasing member **1706**, such as a compression spring **1754**, is configured to bias the handle member **1702** axially (e.g., upward in FIG. **17A**) to be in one of the closed position **1710** or the open position **1712**. In other words, the biasing member **1706** is configured to apply a biasing force (indicated by dashed arrow) to engage rotation locking mechanism **1704** of handle member **1702** with the corresponding rotation locking mechanism **1750** of main frame **602**. One technical benefit is the handle member **1702** may be selectively repositioned between the closed position **1710** and the open position **1712**, and held in the selected position.

[0046] In an embodiment, the handle member **1702** may comprise a round tube **1722** and a grip member **1724**. Grip member **1724** may comprise a round tube, a square tube, or some other element configured to be grasped by a human hand. Grip member **1724** extends radially outward from round tube **1722** (e.g., forming approximately a 90-degree angle). Round tube **1722** is an elongated member having a first end **1730** and an opposing second end **1732**. Round tube **1722** is configured to slide over a round or cylindrical support member (e.g., vertical rail member **614**) of main frame **602**, and rotate in relation to that support member. The first end **1730** of round tube **1722** includes rotation locking mechanism **1704**, and the second end **1732** of round tube **1722** is configured to engage or abut biasing member **1706** (or an intervening member, such as a spacer, bushing, etc.). In FIG. **17C**, for example, the round tube **1722** of handle member **1702** comprises a hollow cylinder **1782** having a diameter larger (e.g., slightly) than the diameter of a (e.g., vertical) support member **1780** of the main frame **602**. As described above, support member **1780** may comprise a vertical rail member **614**, but other types of support members are considered herein. The support member **1780** of main frame **602** may be rotatably fitted into the hollow cylinder **1782** of handle member **1702**. Thus, the round tube **1722** of handle member **1702** and the cylindrical support member **1780** are concentric. In other embodiments, the handle member **1702** may be rotatably fitted into a hollow cylinder of support member **1780**, as an example. Although not

shown, a bushing, bearing, etc., may be disposed between the round tube **1722** of handle member **1702** and the cylindrical support member **1780**. One technical benefit is this configuration allows the handle member **1702** to rotate or pivot in relation to the main frame **602**.

[0047] To transition from the closed position **1710** as in FIG. **17A** to the open position **1712** as in FIG. **17F**, handle member **1702** slides or translates axially against the biasing force applied by the biasing member **1706**, such as shown in FIG. **17B**. This disengages the rotation locking mechanism **1704** of handle member **1702** from the rotation locking mechanism **1750** of main frame **602**. When disengaged, handle member **1702** is free to rotate in relation to the vertical rail member **614** about rotation axis **1770**.

[0048] FIG. **17D** is a magnified view of rotation locking mechanism **1704**. Rotation locking mechanism **1704** may include one or more engagement members **1760** (e.g., a projection(s), notch(es), groove(s), etc.) configured to engage with corresponding engagement members **1762** (e.g., a projection(s), notch(es), groove(s), etc.) of rotation locking mechanism **1750**. For example, rotation locking mechanism **1750** may be fixedly attached to main frame **602** (e.g., vertical rail member **614**). When an engagement member(s) **1760** of rotation locking mechanism **1704** engages with an engagement member(s) **1762** of rotation locking mechanism **1750**, handle member **1702** is locked and unable to rotate. When an engagement member(s) **1760** of rotation locking mechanism **1704** disengages from an engagement member(s) **1762** of rotation locking mechanism **1750**, handle member **1702** is unlocked and free to rotate. Although rotation locking mechanism **1704** is illustrated as a castellated feature on the first end **1730** of handle member **1702** having a plurality of projections separated by slots or notches, rotation locking mechanism **1704** may comprise a single projection or notch as an engagement member **1760** in other embodiments.

[0049] With the rotation locking mechanism **1704** of handle member **1702** disengaged from the rotation locking mechanism **1750** of main frame **602**, handle member **1702** is free to rotate about rotation axis **1770** as illustrated in FIG. **17E**. In an embodiment, handle member **1702** may rotate about rotation axis **1770**, between the closed position **1710** and the open position **1712**, such as in the range of about 75-105 degrees. With handle member **1702** rotated to the open position **1712**, handle member **1702** slides or translates axially (e.g., upward) with the biasing force applied by the biasing member **1706**, such as shown in FIG. **17F**. This engages the rotation locking mechanism **1704** of handle member **1702** with the rotation locking mechanism **1750** of main frame **602**. When engaged, handle member **1702** is locked in the open position **1712**. One technical benefit of a handle member **1702** as illustrated in FIGS. **17A-17F** is the handle members **1702** of wheelchair transport carrier **600** may be opened to lift wheelchair transport carrier **600**, and may be closed for transport so that the shape or size of the wheelchair transport carrier **600** is more compact.

[0050] FIG. **18** is a perspective view of wheelchair transport carrier **600** in an illustrative embodiment. In this view, the ramp **662** is in the open position **1204**, and the lift handle **612** are in the open position **1712**.

[0051] FIGS. **19A-19B** are flow charts illustrating a method **1900** of using a wheelchair transport carrier **600** in an illustrative embodiment. The steps of the flow charts described herein are not all inclusive and may include other steps not shown, and the steps may be performed in an alternative order. For method **1900** in FIG. **19A**, a wheelchair transport carrier **600** is obtained (step **1902**), such as described above, for secure transport of a wheelchair **1402**. It is assumed for this embodiment that the wheelchair transport carrier **600** is placed on the ground, a floor, or some other relatively flat surface. The ramp **662** of the wheelchair transport carrier **600** is lowered to the lowered position (step **1904**), such as by disengaging the ramp latch(es) **1104** and rotating or pivoting the ramp **662** downward via ramp hinge **1106**. With the ramp **662** lowered, a wheelchair **1402** is loaded on the wheelchair support surface **608** of the wheelchair transport carrier **600** via the ramp **662** (step **1906**). The wheelchair **1402** is restrained or anchored to the main frame **602** of wheelchair transport carrier **600** via one or more restraints **1610** (step **1908**). The ramp **662** is raised to the raised position (step **1910**), such as by rotating or pivoting the ramp **662** upward via

ramp hinge **1106** and engaging the ramp latch(es) **1104**. The lift handles **612** are opened or unfolded to the open position (step **1912**), such as by applying a downward force to the handle member **1702** to overcome the biasing force of the biasing member **1706** and disengage the rotation locking mechanism **1704** of handle member **1702** from the corresponding rotation locking mechanism **1750** of main frame **602**, rotating the handle member **1702** about rotation axis **1770** to an open position, and releasing the downward force so that the rotation locking mechanism **1704** of handle member **1702** engages the corresponding rotation locking mechanism **1750** of main frame **602**. With the lift handles **612** opened, the wheelchair transport carrier **600** may be lifted via the lift handles **612** with the wheelchair **1402** restrained (step **1914**), such as onto an aircraft, onto a conveyor, etc. One technical benefit is a wheelchair **1402** may be protected from damage during transport, such as on an aircraft.

[0052] After secure transport of the wheelchair **1402**, the wheelchair **1402** may be unloaded as follows. For method **1900** in FIG. **19B**, the ramp **662** of the wheelchair transport carrier **600** is lowered to the lowered position (step **1916**), such as by disengaging the ramp latch(es) **1104** and rotating or pivoting the ramp **662** downward via ramp hinge **1106**. The restraints **1610** for the wheelchair **1402** are removed (step **1918**), and the wheelchair **1402** is unloaded from the wheelchair transport carrier **600** via the ramp **662** (step **1920**). The ramp **662** is raised to the raised position (step **1920**), such as by rotating or pivoting the ramp **662** upward via ramp hinge **1106** and engaging the ramp latch(es) **1104**. The lift handles **612** are closed or folded to the stowed position (step **1922**), such as by applying a downward force to the handle member **1702** to overcome the biasing force of the biasing member **1706** and disengage the rotation locking mechanism **1704** of handle member **1702** from the corresponding rotation locking mechanism **1750** of main frame **602**, rotating the handle member **1702** about rotation axis **1770** to a stowed position, and releasing the downward force so that the rotation locking mechanism **1704** of handle member **1702** engages the corresponding rotation locking mechanism **1750** of main frame **602**. The wheelchair transport carrier **600** may be stored until a next use.

[0053] Although specific embodiments were described herein, the scope of the disclosure is not limited to those specific embodiments. The scope of the disclosure is defined by the following claims and any equivalents thereof.

Claims

1. A wheelchair transport carrier, comprising: a main frame comprising a support platform defining a wheelchair support surface to support wheels of a wheelchair; a plurality of lift handles to lift the main frame; and a plurality of restraints to anchor the wheelchair to the support platform.
2. The wheelchair transport carrier of claim 1, further comprising: a ramp at a loading end of the main frame configured to transition between a closed position and an open position.
3. The wheelchair transport carrier of claim 2, further comprising: a ramp hinge that attaches the ramp to the main frame at the loading end, wherein the ramp is configured to pivot in relation to the main frame about a horizontal rotation axis via the ramp hinge; and one or more ramp latches configured to secure the ramp in the closed position.
4. The wheelchair transport carrier of claim 1, wherein: the main frame further comprises a plurality of anchor points for the restraints.
5. The wheelchair transport carrier of claim 4, wherein: the restraints comprise automatic retractable tie-downs.
6. The wheelchair transport carrier of claim 4, wherein: the anchor points comprise L-track rails.
7. The wheelchair transport carrier of claim 1, wherein: the main frame is made from aluminum.
8. The wheelchair transport carrier of claim 1, further comprising: one or more wheels mounted on a bottom side of the main frame toward an end of the main frame.
9. The wheelchair transport carrier of claim 1, further comprising: side rails on opposing sides of

the main frame.

10. The wheelchair transport carrier of claim 9, further comprising: a ramp at a loading end of the main frame; and an end rail on an end of the main frame opposite the loading end.

11. The wheelchair transport carrier of claim 9, wherein: the side rails comprise one or more horizontal rail members that span at least a portion of a length of the main frame, and a plurality of vertical rail members that connect the one or more horizontal rail members to the support platform.

12. The wheelchair transport carrier of claim 1, wherein: the lift handles are rotatably coupled to the main frame to transition between a closed position and an open position.

13. The wheelchair transport carrier of claim 12, wherein a lift handle of the lift handles comprises: a handle member rotatably coupled to a vertical support member of the main frame to rotate about a rotation axis; and a rotation locking mechanism configured to prevent rotation of the handle member about the rotation axis.

14. The wheelchair transport carrier of claim 13, wherein the lift handle further comprises: a biasing member configured to apply a biasing force axially in relation to the rotation axis to engage the rotation locking mechanism with a corresponding rotation locking mechanism of the main frame.

15. The wheelchair transport carrier of claim 13, wherein: the handle member comprises a hollow cylinder and the vertical support member of the main frame is rotatably fitted into the hollow cylinder.

16. The wheelchair transport carrier of claim 13, wherein: the rotation locking mechanism comprises at least one engagement member configured to engage with at least one corresponding engagement member of a corresponding rotation locking mechanism of the main frame.

17. The wheelchair transport carrier of claim 1, wherein: a width of the main frame is less than about 31 inches.

18. A wheelchair transport carrier, comprising: a main frame comprising a support platform defining a wheelchair support surface to support wheels of a wheelchair; a ramp at a loading end of the main frame configured to rotate in relation to the main frame between a raised position and a lowered position; a plurality of restraints to anchor the wheelchair to the support platform; and a plurality of lift handles rotatably coupled to the main frame to transition between a stowed position and an open position.

19. The wheelchair transport carrier of claim 18, wherein a lift handle of the lift handles comprises: a handle member rotatably coupled to a vertical support member of the main frame to rotate about a rotation axis; and a rotation locking mechanism configured to lock the handle member in the stowed position or the open position.

20. A method comprising: obtaining a wheelchair transport carrier comprising: a main frame comprising a support platform defining a wheelchair support surface to support wheels of a wheelchair; a ramp at a loading end of the main frame configured to rotate in relation to the main frame between a raised position and a lowered position; a plurality of restraints to anchor the wheelchair to the support platform; and a plurality of lift handles rotatably coupled to the main frame to transition between a stowed position and an open position; lowering the ramp to the lowered position; loading the wheelchair on the wheelchair support surface via the ramp; restraining the wheelchair to the main frame via one or more restraints; raising the ramp to the raised position; opening the lift handles to the open position; and lifting the wheelchair transport carrier via the lift handles with the wheelchair restrained.
